

Workshop Practice Lab:

Workshop Practice Lab is a course designed to provide students with hands-on experience in various workshop processes and techniques. The course typically covers the following topics:

1. **Safety:** Students learn about the importance of safety in the workshop and how to use tools and equipment safely.
2. **Hand Tools:** Students learn about the various hand tools used in the workshop, such as hammers, screwdrivers, pliers, and saws, and how to use them correctly.
3. **Measuring Instruments:** Students learn about the different measuring instruments used in the workshop, such as rulers, calipers, and micrometers, and how to use them accurately.
4. **Machine Tools:** Students learn about the various machine tools used in the workshop, such as lathes, milling machines, and drilling machines, and how to operate them correctly.
5. **Welding:** Students learn about the different types of welding, such as arc welding, gas welding, and TIG welding, and how to perform them safely.
6. **Sheet Metal Work:** Students learn about the various techniques used in sheet metal work, such as cutting, bending, and forming, and how to perform them correctly.
7. **Carpentry:** Students learn about the different types of wood and how to work with them, as well as the various techniques used in carpentry, such as sawing, planing, and sanding.
8. **Fitting:** Students learn about the different techniques used in fitting, such as filing, drilling, and tapping, and how to perform them accurately.

The course provides students with the opportunity to develop practical skills and gain experience in using tools and equipment in a workshop setting. It is an essential course for students pursuing careers in engineering, manufacturing, and related fields.

S. No. Mechanical Workshop Duration

1. **Introduction to Mechanical Workshop:** Mechanical workshop is a place where various mechanical operations are performed using different tools and machines. The duration of a mechanical workshop can vary depending on the scope and objectives of the workshop.
2. **Factors affecting the duration of a mechanical workshop:** The duration of a mechanical workshop can be affected by several factors such as the number of participants, the complexity of the tasks to be performed, the availability of tools and machines, and the level of expertise of the participants.
3. **Typical duration of a mechanical workshop:** A typical mechanical workshop can last anywhere from a few hours to several days. Shorter work-

shops may focus on a specific topic or skill, while longer workshops may cover a broader range of topics and provide more hands-on experience.

4. Importance of planning the duration of a mechanical workshop: It is important to carefully plan the duration of a mechanical workshop to ensure that the objectives of the workshop are met and that the participants have enough time to learn and practice the skills being taught. A well-planned workshop can help participants to gain the most benefit from the experience.
5. Conclusion: The duration of a mechanical workshop can vary depending on several factors. Careful planning and consideration of these factors can help to ensure that the workshop is successful and that the participants gain the most benefit from the experience.

Introduction to Mechanical Workshop Material, Tools and Machines

Mechanical workshop is a place where various materials, tools and machines are used to manufacture, repair or maintain mechanical systems. In this section, we will discuss the following topics:

1. **Materials:** Different types of materials used in mechanical workshop such as metals, plastics, composites, etc.
2. **Tools:** Various hand tools and power tools used in mechanical workshop such as hammers, screwdrivers, drills, etc.
3. **Machines:** Different types of machines used in mechanical workshop such as lathes, milling machines, drilling machines, etc.

Materials

In a mechanical workshop, various materials are used to manufacture, repair or maintain mechanical systems. Some of the commonly used materials are:

- **Metals:** Metals such as steel, aluminum, copper, etc. are widely used in mechanical workshop due to their strength, durability and other properties.
- **Plastics:** Plastics are also used in mechanical workshop due to their lightweight, corrosion resistance and other properties.
- **Composites:** Composites are materials made by combining two or more different materials to achieve desired properties. They are widely used in mechanical workshop due to their high strength-to-weight ratio, corrosion resistance, etc.

Tools

Various hand tools and power tools are used in mechanical workshop to carry out different tasks. Some of the commonly used tools are:

- **Hammers:** Hammers are used to drive nails, fit parts, forge metal, etc.
- **Screwdrivers:** Screwdrivers are used to turn screws with slotted heads.
- **Drills:** Drills are used to make holes in materials.

Machines

Different types of machines are used in mechanical workshop to carry out various tasks. Some of the commonly used machines are:

- **Lathes:** Lathes are used to shape materials by rotating them against a cutting tool.
- **Milling machines:** Milling machines are used to shape materials by removing material using a rotating cutting tool.
- **Drilling machines:** Drilling machines are used to make holes in materials using a rotating cutting tool.

This is a brief introduction to the materials, tools and machines used in a mechanical workshop. In the following sections, we will discuss these topics in more detail.

To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc) used in workshop.

1. **Layout:** The layout of a workshop refers to the arrangement of the equipment, machinery, and workstations in the space. A well-designed layout can improve efficiency, safety, and productivity.
2. **Safety measures:** Safety measures in a workshop include the use of personal protective equipment (PPE) such as safety glasses, gloves, and earplugs; proper ventilation and lighting; and the use of safety guards on machinery. Regular safety training and inspections are also important.
3. **Engineering materials:** Different engineering materials are used in a workshop depending on the specific application. Some common materials include:
 - **Mild steel:** A type of carbon steel with a low carbon content. It is relatively soft and ductile, making it easy to shape and weld. It is commonly used for general purpose structural and mechanical applications.
 - **Medium carbon steel:** A type of carbon steel with a higher carbon content than mild steel. It is stronger and harder, but less ductile. It is commonly used for applications that require strength and toughness, such as shafts, gears, and axles.

- **High carbon steel:** A type of carbon steel with an even higher carbon content. It is very hard and strong, but brittle. It is commonly used for applications that require high strength and wear resistance, such as cutting tools and springs.
- **High speed steel:** A type of tool steel that can withstand high temperatures without losing its hardness. It is commonly used for cutting tools such as drill bits and saw blades.
- **Cast iron:** A group of iron-carbon alloys with a carbon content greater than 2%. It is hard and brittle, but has good wear resistance and can be cast into complex shapes. It is commonly used for engine blocks, pipes, and machine frames.

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.

Fitting, sheet metal, and welding are essential processes in the manufacturing industry. These processes require the use of various tools, equipment, devices, and machines to achieve the desired results. Here is a list of some of the most commonly used tools, equipment, devices, and machines in fitting, sheet metal, and welding sections:

1. **Fitting tools:** These include hand tools such as hammers, chisels, files, hacksaws, and wrenches. These tools are used for cutting, shaping, and fastening metal parts.
2. **Sheet metal tools:** These include tools such as shears, punches, and bending brakes. These tools are used for cutting, shaping, and bending sheet metal.
3. **Welding equipment:** This includes equipment such as welding machines, welding torches, and protective gear. Welding equipment is used to join metal parts together using heat and pressure.
4. **Measuring and marking tools:** These include tools such as rulers, calipers, and scribes. These tools are used to measure and mark metal parts for cutting and shaping.
5. **Machines:** These include machines such as lathes, milling machines, and drill presses. These machines are used for cutting, shaping, and drilling metal parts.

It is important to use the right tools, equipment, devices, and machines for each process to achieve the desired results. Proper training and safety precautions should also be followed when using these tools and equipment.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

1. **Vernier Calliper:** The least count of a vernier calliper is the difference between the value of one main scale division and the value of one vernier

scale division. To calculate the least count, divide the value of one main scale division by the total number of vernier scale divisions.

2. **Vernier Height Gauge:** The least count of a vernier height gauge is calculated in the same way as that of a vernier calliper. Divide the value of one main scale division by the total number of vernier scale divisions.
3. **Micrometer (Screw Gauge):** The least count of a micrometer is the smallest distance that can be measured with the instrument. It is equal to the pitch of the screw, which is the distance moved by the spindle per revolution, divided by the total number of divisions on the thimble.

To take different readings over given metallic pieces using these instruments, follow these steps:

1. **Vernier Calliper:** Place the metallic piece between the jaws of the vernier calliper and close the jaws until they touch the piece. Read the main scale and add the vernier scale reading to obtain the final measurement.
2. **Vernier Height Gauge:** Place the metallic piece on a flat surface and lower the vernier height gauge until it touches the piece. Read the main scale and add the vernier scale reading to obtain the final measurement.
3. **Micrometer (Screw Gauge):** Place the metallic piece between the anvil and the spindle of the micrometer and rotate the thimble until the piece is held firmly. Read the main scale and add the thimble reading to obtain the final measurement.

Machine Shop

A machine shop is a facility where machining is performed. Machining is the process of removing material from a workpiece to create a desired shape or size. This is typically done using machine tools such as lathes, milling machines, and drill presses. Here are some key points to know about machine shops:

1. **Equipment:** Machine shops typically contain a variety of equipment, including lathes, milling machines, drill presses, grinders, and saws. These machines are used to cut, shape, and form metal and other materials.
2. **Materials:** Machine shops work with a wide range of materials, including metals, plastics, and composites. The choice of material depends on the desired properties of the final product, such as strength, durability, and resistance to heat and corrosion.
3. **Processes:** There are many different processes that can be performed in a machine shop, including turning, milling, drilling, and grinding. These processes are used to remove material from a workpiece to create a desired shape or size.

4. **Safety:** Safety is a top priority in machine shops. Workers must follow strict safety protocols to prevent accidents and injuries. This includes wearing protective equipment, such as safety glasses and gloves, and following proper procedures when operating machinery.
5. **Training:** Working in a machine shop requires specialized training and knowledge. Workers must be skilled in operating and maintaining the equipment, as well as understanding the properties of the materials they are working with.
6. **Precision:** Machine shops are capable of producing parts with a high degree of precision and accuracy. This is important for applications where tight tolerances and exact specifications are required.
7. **Customization:** Machine shops can produce custom parts to meet the specific needs of their customers. This can include creating one-of-a-kind prototypes or producing large quantities of a specific part.

In summary, a machine shop is a facility where machining is performed to create parts with a desired shape or size. Machine shops contain a variety of equipment and work with a wide range of materials. Safety, training, and precision are all important aspects of working in a machine shop. Machine shops are capable of producing custom parts to meet the specific needs of their customers.

Demonstration of working, construction and accessories for Lathe machine Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.

A lathe is a machine tool that rotates a workpiece about an axis of rotation to perform various operations such as cutting, sanding, knurling, drilling, deformation, facing, and turning. The lathe is one of the oldest and most important machine tools.

The main parts of a lathe machine are: - Headstock: It is the part of the lathe machine that holds and rotates the workpiece. - Tailstock: It is the part of the lathe machine that supports the other end of the workpiece. - Bed: It is the base of the lathe machine on which the headstock and tailstock are mounted. - Carriage: It is the part of the lathe machine that moves the cutting tool along the workpiece. - Cross slide: It is the part of the lathe machine that moves the cutting tool perpendicular to the workpiece. - Compound rest: It is the part of the lathe machine that holds the cutting tool and can be rotated to make angular cuts. - Tool post: It is the part of the lathe machine that holds the cutting tool.

The main operations that can be performed on a lathe machine are: - Facing: It is the operation of machining the ends of a workpiece to produce a flat surface. - Plane Turning: It is the operation of machining a cylindrical surface on a workpiece. - Step Turning: It is the operation of machining different diameters on a workpiece. - Taper Turning: It is the operation of machining a conical

surface on a workpiece. - Threading: It is the operation of cutting threads on a workpiece. - Knurling: It is the operation of producing a rough surface on a workpiece. - Parting: It is the operation of cutting a workpiece into two or more pieces.

These are the basic operations that can be performed on a lathe machine. There are many other operations and accessories that can be used to perform more complex tasks. It is important to understand the construction and working of the lathe machine to be able to perform these operations effectively.

Fitting Shop

Fitting shop is a workshop where the process of fitting, assembling, and finishing of machine parts or components takes place. The following points provide an overview of the fitting shop:

1. **Tools and equipment:** The fitting shop is equipped with various hand tools and machines such as files, hammers, chisels, hacksaws, vices, and drilling machines, among others. These tools are used to perform operations such as cutting, shaping, and finishing of metal components.
2. **Materials:** The materials commonly used in the fitting shop include metals such as steel, iron, brass, and aluminum. These materials are available in various shapes and sizes, such as rods, bars, sheets, and plates.
3. **Operations:** The operations performed in the fitting shop include marking, cutting, filing, drilling, tapping, and reaming, among others. These operations are carried out to achieve the desired shape, size, and finish of the components.
4. **Safety:** Safety is an important aspect of the fitting shop. Workers are required to wear protective gear such as gloves, goggles, and aprons to prevent injuries. Additionally, the tools and machines must be used in accordance with the safety guidelines to prevent accidents.
5. **Applications:** The fitting shop plays a crucial role in the manufacturing industry. The components produced in the fitting shop are used in the assembly of various machines and equipment. Additionally, the fitting shop also undertakes repair and maintenance work.

In summary, the fitting shop is a workshop where various operations are carried out to produce, assemble, and finish machine components. It is equipped with various tools and machines and plays a crucial role in the manufacturing industry. Safety is an important aspect of the fitting shop, and workers are required to follow safety guidelines to prevent accidents. The duration of the fitting shop is 3 hours.

1. Practice marking operations

Marking operations are an essential part of many manufacturing processes. They are used to identify, label, and track products, components, and materials. Here are some points to consider when practicing marking operations:

1. **Choose the right marking method:** There are several marking methods available, including inkjet, laser, and dot peen. Each method has its advantages and disadvantages, so it's important to choose the right one for your application.
2. **Ensure proper placement:** The placement of the mark is important for readability and accuracy. Make sure the mark is placed in a location that is easily visible and won't be obscured by other components or materials.
3. **Use the correct marking material:** The marking material should be chosen based on the surface being marked and the durability requirements of the mark. Some common marking materials include ink, paint, and metal.
4. **Follow safety guidelines:** Marking operations can involve the use of hazardous materials and equipment. Be sure to follow all safety guidelines and wear appropriate personal protective equipment.
5. **Maintain equipment:** Proper maintenance of marking equipment is essential for ensuring consistent and accurate marks. Follow the manufacturer's recommended maintenance schedule and keep the equipment clean and in good working order.
6. **Verify the mark:** After the mark has been applied, it's important to verify that it is legible, accurate, and meets the required specifications. This can be done through visual inspection or by using specialized equipment.

By following these guidelines, you can improve your practice of marking operations and ensure that your marks are accurate, legible, and durable.

2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

The preparation of U or V -Shape Male Female Work piece involves several steps, including filing, sawing, drilling, and grinding. These steps are necessary to achieve the desired shape and fit of the work piece.

1. **Filing:** Filing is the process of removing material from the work piece using a file. This is done to smooth out rough edges and to shape the work piece to the desired form.
2. **Sawing:** Sawing is the process of cutting the work piece to the desired size and shape using a saw. This can be done using a hand saw or a power saw, depending on the size and thickness of the work piece.
3. **Drilling:** Drilling is the process of creating holes in the work piece using a drill. This is necessary for attaching the work piece to other components

or for creating openings for screws or bolts.

4. **Grinding:** Grinding is the process of removing material from the work piece using an abrasive wheel or disc. This is done to smooth out rough surfaces and to achieve a high level of precision in the final shape of the work piece.

By following these steps, a U or V -Shape Male Female Work piece can be prepared to the desired specifications. It is important to use the appropriate tools and techniques for each step to ensure a high-quality result.

4 Carpentry Shop 3 Hrs

Carpentry is a skilled trade and a craft in which the primary work performed is the cutting, shaping, and installation of building materials during the construction of buildings, ships, timber bridges, concrete formwork, etc. Carpenters traditionally worked with natural wood and did the rougher work such as framing, but today many other materials are also used and sometimes the finer trades of cabinetmaking and furniture building are considered carpentry.

A carpentry shop is a workshop where carpenters work. It is typically equipped with tools and machinery for cutting, shaping, and joining wood. Some common tools and machinery found in a carpentry shop include:

1. Saws: Used for cutting wood to size. Some common types of saws found in a carpentry shop include hand saws, circular saws, and jigsaws.
2. Planes: Used for smoothing and shaping wood. Some common types of planes found in a carpentry shop include hand planes, block planes, and jointer planes.
3. Chisels: Used for cutting and shaping wood. Some common types of chisels found in a carpentry shop include bevel-edge chisels, mortise chisels, and paring chisels.
4. Hammers: Used for driving nails and other fasteners. Some common types of hammers found in a carpentry shop include claw hammers, framing hammers, and ball-peen hammers.
5. Drills: Used for making holes in wood. Some common types of drills found in a carpentry shop include hand drills, power drills, and drill presses.

A carpentry shop may also have specialized machinery such as a table saw, band saw, or lathe for more advanced woodworking tasks.

In a 3-hour session at a carpentry shop, one could expect to learn and practice basic carpentry skills such as measuring, cutting, and joining wood. One could also expect to learn about the different types of wood and their properties, as well as how to select the right wood for a particular project. Additionally, one could expect to learn about the safe and proper use of tools and machinery in the carpentry shop.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

Carpentry is a skilled trade that involves the cutting, shaping, and installation of building materials during the construction of buildings, ships, timber bridges, concrete formwork, etc. Carpenters use various tools and equipment to carry out their work. Some of the common tools and equipment used in carpentry include:

1. **Saws:** Saws are used to cut wood to the desired size and shape. There are different types of saws, including hand saws, circular saws, and jigsaws.
2. **Hammers:** Hammers are used to drive nails into wood. There are different types of hammers, including claw hammers and framing hammers.
3. **Chisels:** Chisels are used to shape wood by removing small pieces of material. There are different types of chisels, including bevel-edge chisels and mortise chisels.
4. **Planes:** Planes are used to smooth and shape wood. There are different types of planes, including block planes and smoothing planes.
5. **Squares:** Squares are used to ensure that angles are accurate. There are different types of squares, including combination squares and framing squares.
6. **Measuring tools:** Measuring tools are used to ensure that cuts are accurate. There are different types of measuring tools, including tape measures and rulers.

Carpenters use different joints to join pieces of wood together. Some of the common joints used in carpentry include:

1. **Cross Half Lap Joint:** A cross half lap joint is made by removing half of the thickness of two pieces of wood and fitting them together. This joint is strong and is often used in framing.
2. **Half Lap Dovetail Joint:** A half lap dovetail joint is similar to a cross half lap joint, but the ends of the wood are cut in a dovetail shape. This joint is strong and is often used in furniture making.
3. **Mortise and Tenon Joint:** A mortise and tenon joint is made by cutting a mortise (hole) in one piece of wood and a tenon (tongue) on the other piece of wood. The tenon is then inserted into the mortise and the joint is secured with glue or a peg. This joint is strong and is often used in furniture making.

In conclusion, carpentry involves the use of various tools and equipment to cut, shape, and install building materials. Carpenters use different joints to join pieces of wood together, including cross half lap joints, half lap dovetail joints,

and mortise and tenon joints. These joints are strong and are used in various applications, including framing and furniture making.

5 Welding Shop 6 Hrs

Welding is a fabrication process that joins two or more pieces of metal or thermoplastics by heating the surfaces to their melting points and applying pressure to fuse them together. Welding is used in a variety of industries, including construction, manufacturing, and repair.

Here are some key points to consider when setting up a welding shop for 6 hours of operation:

1. **Equipment:** Ensure that you have all the necessary equipment, including welding machines, protective gear, and tools. Make sure that the equipment is in good working condition and is suitable for the type of welding you will be doing.
2. **Safety:** Welding can be dangerous, so it is important to follow all safety guidelines and regulations. Make sure that the welding area is well-ventilated, and that all workers are trained in proper safety procedures.
3. **Materials:** Make sure that you have all the materials you will need for the welding job, including the metal or thermoplastic pieces to be joined, filler material, and shielding gas.
4. **Preparation:** Proper preparation is key to a successful welding job. Make sure that the pieces to be joined are clean and free of any contaminants that could affect the quality of the weld.
5. **Time management:** With only 6 hours of operation, it is important to manage your time effectively. Plan out the welding job in advance, and make sure that all workers are aware of their responsibilities and the timeline for completion.

By following these guidelines, you can set up a successful welding shop for 6 hours of operation. Remember to always prioritize safety and quality in your work.

Introduction to BI standards and reading of welding drawings. Practice of Making following operations Butt Joint Lap Joint TIG Welding MIG Welding

BI standards, or Business Intelligence standards, are a set of guidelines and best practices for the design, development, and implementation of BI systems. These standards help ensure that BI systems are effective, efficient, and provide accurate and reliable information to support business decision-making.

Reading welding drawings is an essential skill for anyone involved in the welding industry. Welding drawings provide detailed information about the design and specifications of a welded joint, including the type of joint, the materials to be used, and the welding process to be employed.

Butt Joint: A butt joint is a type of joint where two pieces of material are joined end to end. This type of joint is commonly used in welding to join two pieces of metal together.

Lap Joint: A lap joint is a type of joint where two pieces of material are overlapped and joined together. This type of joint is commonly used in welding to join two pieces of metal together.

TIG Welding: TIG welding, or Tungsten Inert Gas welding, is a type of welding process that uses a non-consumable tungsten electrode to produce the weld. This type of welding is commonly used for welding thin materials and for welding in tight spaces.

MIG Welding: MIG welding, or Metal Inert Gas welding, is a type of welding process that uses a consumable wire electrode to produce the weld. This type of welding is commonly used for welding thicker materials and for welding in open spaces.

6 Moulding and Casting Shop 6 Hrs

Moulding and casting are manufacturing processes used to produce complex shapes. These processes involve pouring molten metal into a mould, which contains a hollow cavity of the desired shape, and then allowing the metal to cool and solidify. The solidified part is then removed from the mould to complete the process.

1. **Moulding** is the process of manufacturing by shaping liquid or pliable raw material using a rigid frame called a mould or matrix.
2. **Casting** is a manufacturing process in which a liquid material is usually poured into a mould, which contains a hollow cavity of the desired shape, and then allowed to solidify.
3. The most common materials used in casting are metals, but other materials such as plastics, glass, and concrete can also be cast.
4. The moulding and casting process can be used to produce a wide variety of products, including engine blocks, pipes, and decorative items.
5. There are several different types of moulding and casting processes, including sand casting, investment casting, and die casting.
6. Each process has its own advantages and disadvantages, and the choice of process will depend on factors such as the desired shape, size, and material of the final product.

Introduction to Patterns, Pattern Allowances, Ingredients of Moulding Sand and Melting Furnaces. Foundry Tools and their Purposes. Demo of Mould Preparation and Aluminum Casting Practice – Study and Preparation of Mould for Plastic

1. **Patterns:** A pattern is a model or replica of the object to be cast. It is used to create the cavity in the mould where the molten metal will be poured to form the casting.
2. **Pattern Allowances:** Pattern allowances are small adjustments made to the pattern to compensate for changes that occur during the casting process. These allowances include shrinkage allowance, machining allowance, draft allowance, and distortion allowance.
3. **Ingredients of Moulding Sand:** Moulding sand is a mixture of sand, clay, and water. The sand provides the structure for the mould, the clay binds the sand together, and the water helps to make the sand more workable.
4. **Melting Furnaces:** Melting furnaces are used to melt the metal that will be used for casting. There are several types of melting furnaces, including cupola furnaces, electric arc furnaces, and induction furnaces.
5. **Foundry Tools:** Foundry tools are used in the casting process to shape, form, and finish the casting. These tools include rammers, trowels, sprue cutters, and gate cutters.
6. **Demo of Mould Preparation and Aluminum Casting Practice:** This involves the preparation of the mould using the pattern and the pouring of molten aluminum into the mould to create the casting.
7. **Study and Preparation of Mould for Plastic:** This involves the study of the properties of plastic and the preparation of the mould to cast plastic parts.

7 CNC Shop 6 Hrs

1. CNC stands for Computer Numerical Control.
2. A CNC shop is a workshop where CNC machines are used to manufacture parts and components.
3. CNC machines are controlled by computer programs, which allow for precise and accurate cutting, shaping, and drilling of materials.
4. A 6-hour shift in a CNC shop would involve operating and monitoring the CNC machines, loading and unloading materials, and performing quality checks on finished products.
5. CNC shops can produce a wide range of parts and components, from small, intricate pieces to large, complex structures.
6. CNC technology has revolutionized the manufacturing industry, allowing for faster production times and higher levels of precision and accuracy.
7. Working in a CNC shop requires technical knowledge and skills, as well as the ability to read and interpret technical drawings and specifications.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

CNC (Computer Numerical Control) machines are automated milling devices that make industrial components without direct human assistance. They use coded instructions that are sent to an internal computer, which allows factories to fabricate parts accurately and quickly.

The main parts of a CNC machine are: 1. **Input Devices:** These are devices that are used to input the part program into a CNC machine. 2. **Machine Control Unit (MCU):** The machine control unit is called the heart of the CNC machine. 3. **Machine Tool:** A CNC machine tool always has a sliding table and a spindle to control the position and speed. 4. **Driving System:** This system is responsible for the movement of the different machine axes. 5. **Feedback System:** This system is responsible for ensuring the correct position of the machine axes. 6. **Display Unit:** This unit displays the information to the operator.

CNC machines can be used to perform different operations on metal components, such as milling, drilling, and cutting. The specific operations that can be performed depend on the type of CNC machine and the accessories that are used with it.

8 To prepare a product using 3D printing 3 Hrs

1. **Design the 3D model:** The first step in preparing a product using 3D printing is to design a 3D model of the product using computer-aided design (CAD) software. This model will serve as the blueprint for the 3D printer to create the physical product.
2. **Choose the right 3D printing technology:** There are several 3D printing technologies available, each with its own advantages and disadvantages. The most common technologies are Fused Deposition Modeling (FDM), Stereolithography (SLA), and Selective Laser Sintering (SLS). Choose the technology that best suits the requirements of your product.
3. **Select the appropriate material:** 3D printing can be done with a variety of materials, including plastics, metals, and ceramics. Choose the material that best suits the requirements of your product, taking into account factors such as strength, flexibility, and durability.
4. **Prepare the 3D printer:** Before starting the printing process, make sure that the 3D printer is properly set up and calibrated. This includes loading the material, setting the printing parameters, and ensuring that the build platform is level.
5. **Start the printing process:** Once the 3D printer is prepared, start the

printing process. The 3D printer will read the 3D model and build the product layer by layer. The printing time will vary depending on the size and complexity of the product.

6. **Post-processing:** After the printing process is complete, the product may require some post-processing. This can include removing support structures, sanding, polishing, or painting. The post-processing steps will vary depending on the 3D printing technology and the material used.
7. **Quality control:** Finally, perform a quality control check to ensure that the product meets the desired specifications. This can include checking the dimensions, strength, and appearance of the product.

In summary, preparing a product using 3D printing involves designing a 3D model, choosing the right 3D printing technology and material, preparing the 3D printer, starting the printing process, performing post-processing, and conducting a quality control check. The entire process can take around 3 hours, depending on the size and complexity of the product.

Total 33 Hrs

- The topic of “Total 33 Hrs” is not clear. Could you please provide more information or context about this topic?
- Is “Total 33 Hrs” related to a specific subject or event?

Course Outcome:

A course outcome is a statement that describes the knowledge, skills, and abilities that students are expected to have acquired by the end of a course. Course outcomes are used to:

1. Define the goals and objectives of a course.
2. Communicate expectations to students.
3. Guide the development of course content and assessments.
4. Evaluate the effectiveness of a course in achieving its goals.

Course outcomes should be specific, measurable, achievable, relevant, and time-bound. They should be aligned with the overall program outcomes and the institution’s mission and goals. Course outcomes should be regularly reviewed and updated to ensure that they remain relevant and effective in achieving the desired learning outcomes for students.

The students will be able to:

CO1. Use various engineering materials, tools, machines and measuring equipments. (Blooms Level: K3) - Students will learn about the properties and applications of different engineering materials. - They will be trained to use various tools, machines and measuring equipments in an engineering setting.

CO2. Perform machine operations in lathe and CNC machine. (Blooms Level: K3) - Students will learn to operate lathe and CNC machines. - They will be able to perform various machining operations such as turning, facing, drilling, and threading.

CO3. Perform manufacturing operations on components in fitting and carpentry shop. (Blooms Level: K3) - Students will learn to perform various manufacturing operations in fitting and carpentry shops. - They will be able to produce components using techniques such as sawing, chiseling, and planing.

CO4. Perform operations in welding, moulding, casting and gas cutting. (Blooms Level: K3) - Students will learn to perform various operations in welding, moulding, casting and gas cutting. - They will be able to produce components using techniques such as arc welding, sand casting, and oxy-fuel cutting.

CO5. Fabricate a job by 3D printing manufacturing technique. (Blooms Level: K3) - Students will learn to use 3D printing technology to fabricate components. - They will be able to produce components using techniques such as fused deposition modeling and stereolithography.

Reference Books:

- Reference books are a type of non-fiction book that provides factual information on a particular subject.
- They are often used for research and academic purposes.
- Some common types of reference books include dictionaries, encyclopedias, almanacs, and atlases.
- Reference books are typically organized in a way that makes it easy to find the information you need.
- They often include indexes, tables of contents, and other tools to help you navigate the content.
- Reference books are usually written by experts in the field and are considered reliable sources of information.
- They are often updated regularly to ensure that the information they contain is accurate and up-to-date.
- Some popular reference book publishers include Oxford University Press, Cambridge University Press, and Merriam-Webster.
- Reference books can be found in libraries, bookstores, and online.
- When using a reference book for research, it is important to evaluate the credibility of the source and to cite it properly in your work.

1. Workshop Practice, H S Bawa, McGraw Hill

Workshop Practice is a book written by H S Bawa and published by McGraw Hill. It is a comprehensive guide to the various aspects of workshop practice, including:

- Introduction to workshop practice
- Safety in the workshop
- Hand tools and their uses
- Measuring instruments and their uses
- Marking and layout tools
- Cutting tools and their uses
- Machine tools and their operations
- Joining processes
- Finishing processes
- Maintenance of tools and equipment

This book is a valuable resource for students and professionals alike, providing detailed information and practical guidance on the various aspects of workshop practice. It is written in a clear and concise manner, making it easy to understand and follow. The book is well-organized and includes numerous illustrations and diagrams to aid in understanding the concepts presented.

2. Mechanical Workshop Practice, K C John, PHI

- “Mechanical Workshop Practice” is a book written by K.C. John and published by PHI Learning in 2010 .
- The book describes the general principles of different workshop processes such as primary and secondary shaping processes, metal joining methods, surface finishing, and heat treatment .
- The book has 240 pages and has received 4 reviews on Google Books .
- The ISBN-10 of the book is 812034166X and the ISBN-13 is 9788120341661 .

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications

- Workshop Practice is a two-volume book set written by HazraChoudhary and published by Media Promoters and Publications.
- The books provide a comprehensive guide to workshop practices and techniques.
- Volume 1 covers topics such as hand tools, measuring instruments, marking and layout, fitting, sheet metal work, and carpentry.
- Volume 2 covers topics such as welding, smithy and forging, foundry, turning, and machine tools.
- The books are written in a clear and concise manner, making them suitable for students and professionals alike.
- The books include numerous diagrams and illustrations to aid in understanding the concepts presented.
- Overall, Workshop Practice Vol 1 and Vol 2 by HazraChoudhary provide a valuable resource for anyone interested in workshop practices and techniques.

CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication

- CNC stands for Computer Numerical Control.
- CNC machines are used for manufacturing processes that require high precision and repeatability.
- CNC machines are controlled by a computer program that contains instructions for the machine's movements and actions.
- CNC programming involves creating the code that the machine will use to perform the desired operations.
- CNC programming can be done using a variety of software tools, including CAD (Computer-Aided Design) and CAM (Computer-Aided Manufacturing) software.
- CNC machines can be used for a wide range of manufacturing processes, including milling, turning, drilling, and grinding.
- CNC machines can be used to produce parts with complex geometries and tight tolerances.
- CNC machines can be used to produce both one-off custom parts and large quantities of identical parts.
- CNC machines can be used to produce parts from a wide range of materials, including metals, plastics, and composites.
- CNC machines can be used to produce parts for a wide range of industries, including aerospace, automotive, and medical.